

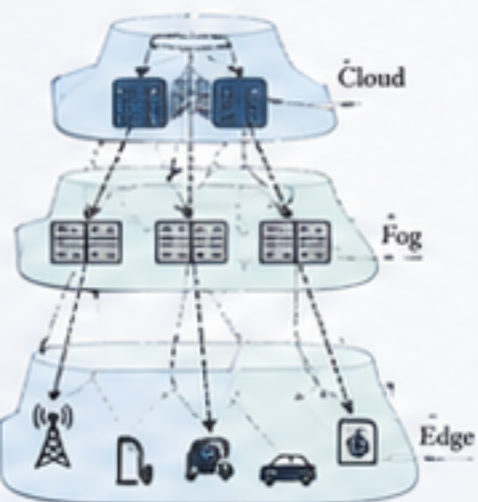
CAGE-S: Legality-First Online Scheduling

Input at timestamp t

Event batch E_t

$e_{t,1}$
 $e_{t,2}$
 $\vdots$
 e_{t,B_t}

Cluster snapshot



1 Policy gate

Is the node permitted?

Policy mask $L(s, \cdot)$

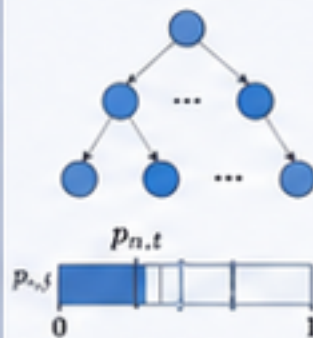
	n_1	n_2	...	n_M
s_1	✓	✓	...	✓
s_2	✓	✓	✗	✓
$\vdots$				
s_Q	✓	✗	✗	✗

Illegal nodes are excluded (hard filter)

2 Temporal risk

Likely to cross in next interval?

HGB model



Accept if $p_{n,t} \leq \theta$

3 Fair shortlisting

Avoid concentration

Order by

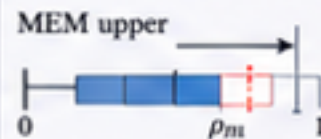
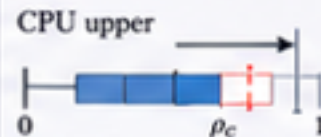
- 1 $H_t(n) \uparrow$
- 2 $g_{n,t} \uparrow$
- 3 $\text{id}(n)$

Keep top k candidates

4 Q95 guardrail

Empirical headroom before commit?

Q95 projection



Accept if $g_{n,t} \leq 1$

Decision & reservation

Selected node

n_e^*



Reserve

$$U_t(n_e^*) \leftarrow U_t(n_e^*) + 1$$



Update histories: $H_t(n_e^*) \leftarrow H_t(n_e^*) + 1$
(Historical counts for fairness across time)